

TECHNICAL SERVICE BULLETIN	
TSB No: 26-1041	Date: July 2026
Group: Powertrain / Thermal Management	Subject: Charge Air Cooler (CAC) Thermal Saturation
Engine: 2.7L EcoBoost V6	Affected Vehicles: 2018 Ford F-150 Status: Technical Advisory

CONDITION

Some vehicles operating in high-ambient temperature regions (90°F+) may exhibit a perceived loss of top-end power, delayed throttle response, or sluggish acceleration during repeated Wide Open Throttle (WOT) applications or heavy towing. The Malfunction Indicator Lamp (MIL) will NOT be illuminated, and no Diagnostic Trouble Codes (DTCs) will be stored in the Powertrain Control Module (PCM).

TECHNICAL CAUSE

Engineering telemetry indicates the factory air-to-air Charge Air Cooler (CAC) can reach thermal saturation (“heat soak”) under sustained maximum boost delivery (211 kPa / 16.2 psi). When post-intercooler Intake Air Temperatures (IAT2) exceed 45°C, the PCM actively retards ignition timing to prevent pre-ignition/detonation, resulting in an intentional reduction of torque output. This condition is often exacerbated by debris blocking the CAC fins or by mechanical binding of the Active Grille Shutter (AGS).

DIAGNOSTIC & SERVICE PROCEDURE

Do not replace the turbochargers, wastegate actuators, or high-pressure fuel pumps for this condition. Proceed with the following thermal diagnostic protocol:

Step 1: Visual and Mechanical Inspection

- 1. Inspect the front bumper lower intake aperture for physical obstructions (leaves, plastic bags, mud).
- 2. Using FORScan or equivalent diagnostic interface, command the Active Grille Shutters (AGS) from 0% to 100%. Verify smooth, unobstructed travel of the louvers.

Step 2: Dynamic Telemetry Validation

- 1. Connect a high-speed OBD interface and monitor the following PIDs: RPM, MAP, IAT2, WGATE_PRES, SPARKADV.
- 2. Bring the engine to normal operating temperature (ECT > 195°F).
- 3. Perform a 3rd-gear WOT sweep from 2,500 to 5,000 RPM in a safe environment.

4. Analyze the data log: If MAP reaches >200 kPa but IAT2 climbs rapidly past 45°C and SPARKADV drops, the CAC is experiencing thermal saturation.

Step 3: Corrective Action

1. If the AGS is binding, clear the obstruction or replace the shutter assembly.
2. Clean the exterior fins of the CAC using low-pressure water and compressed air (max 30 psi) to restore thermal transfer efficiency.

CORRECTIVE ACTION & VERIFICATION

Corrective action was completed per Step 3 of the diagnostic procedure above (obstruction clearance / AGS service and CAC fin cleaning). Verify the repair using the protocol below before returning the vehicle to service.

Perform a secondary WOT sweep. Ensure IAT2 recovery times have improved, and spark advance remains steady under peak boost pressure.